

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks: 40

Annexure A

CASE STUDY

Code of Conduct:

I K Akshitha (192312527) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Akshitha

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Vehicle Fault Diagnosis

(a) Propositional Logic

let:

- E = Engine warning light
- N = Unusual Noise
- O = low oil pressure
- H = overheating.
- F = Engine Failure Risk
- L = Oil Leak
- C = Coolant System Failure.

Rules:

1. $E \wedge N \rightarrow F$
2. $E \wedge O \rightarrow L$
3. $N \wedge H \rightarrow C$

Facts:

- E = true
- N = true
- H = true.

(b) Forward chaining

Step 1:

$E = \text{true}, N = \text{true}$

Using Rule 1:

$E \wedge N \rightarrow F \quad \therefore F = \text{true}$

→ possible Engine Failure Risk.

Step 2: Rule 2 needs E and O.

O is not given.

→ oil leak cannot be derived.

Step 3:

$N = \text{True}, H = \text{True}.$

Using Rule 3: $N \wedge H \rightarrow c$

$\therefore c = \text{True} \rightarrow \text{possible coolant system failure}.$

(c) Backward chaining

Goal:

coolant system failure (c)

Find the rule that gives c:

$N \wedge H \rightarrow c$

$N = \text{Unusual Noise}$

$H = \text{Overheating}.$

Both are already facts.

② Hospital Routing Agent

(a) unification

Rule 1: $\text{patient}(x) \wedge \text{critical condition}(x) \rightarrow \text{priority corridor}(x)$

Facts:

- $\text{patient}(\text{patient A})$
- $\text{critical condition}(\text{patient A}).$

Match:

$x = \text{patient A} \quad \therefore \quad \theta = \{x \mid \text{patient A}\}$

So, priority corridor (patient A). is derived.

(b) Resolution

Rule 1: $\text{patient}(x) \wedge \text{critical condition}(x) \rightarrow \text{priority corridor}(x).$

Rule 2: $\text{priority corridor}(x) \rightarrow \text{corridor cleared}(x).$

Rule 3: $\text{patient}(x) \wedge \text{waiting Behind}(x, y) \wedge \text{corridor cleared}(y) \rightarrow \text{Hold}(x).$

Facts:-

- Patient (A)
- Critical condition (A)
- Patient (B)
- Waiting Behind (B, A)

(c) Two critical patients

The current KB can identify two critical patients because Rule 1 uses $\forall x$.

- which patient should go first.
- How robots avoid collision.
- How to handle two patients using the same corridor.

③ Flight scheduling

(a) CNF conversion (P1)

Stormwarning $\rightarrow$ Flight Delay Needed.

CNF: $\rightarrow$ Stormwarning $\vee$ Flight Delay Needed.

(P2)

Flight Delay Needed $\wedge$ International Route $\rightarrow$ Notify Passengers.

CNF: $-$ Flight Delay Needed $\vee -$ International Route $\vee$ Notify passengers.

(P3)

$-$ Notify passengers $\wedge$ International Route $\rightarrow$ Compliance Risk.

CNF: Notify passengers $\vee -$ International Route $\vee$ Compliance Risk.

(b) Resolution Refutation

Notify passengers.

Negate goal:

$\rightarrow$ Notify passengers.

(c) Remove Stormwarning

- Flight Delay needed cannot be derived.
- Notify passengers cannot be derived.
- International Route is still true.

④ Cave Exploration

(a) Belief State.

1. Draft [1,3]
2. Glow [1,1]
3. Sparkle [2,3].

Rule 1 : Draft [1,3] $\rightarrow$ Pit adjacent to [1,3].

Rule 2 : Glow [1,1] $\rightarrow$ Lava vent adjacent to [1,1].

Rule 3 : Sparkle [2,3] $\rightarrow$ Treasure [2,3].

(b) Forward chaining.

Step 1:

Draft [1,3].

$\downarrow$

Pit is adjacent to [1,3].

[1,2], [2,3] and [1,4] if present.

(c) Is the path safe?

Path:

[1,1] $\rightarrow$ [1,3] $\rightarrow$ [2,3]

- [1,1] has Glow $\rightarrow$ Lava vent may be nearby.
- [1,3] has Draft $\rightarrow$ Pit may be nearby.
- [2,3] has Treasure.

The path cannot be confirmed as safe.

More information is needed to prove that all three cells are safe.